

AutomatiCubEx: A Visual Step-by-Step Guide for Extended Emission Extraction

This guide illustrates a standard workflow to extract faint extended emission around luminous sources such as quasars and galaxies using AutomatiCubEx. The software combines an interactive GUI with CubEx/CubExtractor tools, allowing users to visualize datacubes, inspect spectra, apply processing routines, and generate scientific products such as masks and moment maps.

1. Understanding the GUI Interface

The main AutomatiCubEx GUI is organized into three main sections: the image viewer (top), the spectrum viewer (bottom), and the control panel (right).

The upper panel displays the current spatial image extracted from the datacube. This can correspond either to a single spectral slice or to a mean/median image generated from a selected spectral interval. The image can be explored interactively using the matplotlib toolbar shown in red. These controls allow the user to:

- move across the image,
- zoom in and out,
- reset the visualization,
- save the current figure,
- inspect coordinates interactively.

The selected spatial position is marked with a black “+” symbol at coordinates (x,y). The spectrum shown in the lower panel is automatically extracted from this position. The user can also define a circular aperture radius, allowing the spectrum to be averaged over multiple spatial pixels instead of a single spaxel.

The lower panel displays the extracted spectrum corresponding to the selected spatial position or aperture. The user can zoom only along the wavelength axis and this zoom level remains fixed while exploring different spatial positions. A spectral region can be selected directly on the spectrum using the mouse. The selected interval is highlighted in blue between z_{min} and z_{max} (red vertical lines). Once selected, the image shown in the upper panel is immediately updated using either the mean or median flux within the chosen spectral range. This functionality is particularly useful for identifying faint extended emission lines.

The spectral slider at the bottom allows the user to move interactively through individual spectral slices of the datacube. The current spectral channel z and the corresponding wavelength are displayed in the information panel on the right.

The right control panel contains all the main workflow controls:

- “Load FITS cube” loads a new datacube,
- “+ another” loads additional files simultaneously,
- “load current” reloads the currently selected file,

- “remove current data” removes the current file from memory.

Below these controls, the user can specify:

- the CubEx installation path,
- the CSV registry table used to save the history of all applied procedures and parameters.

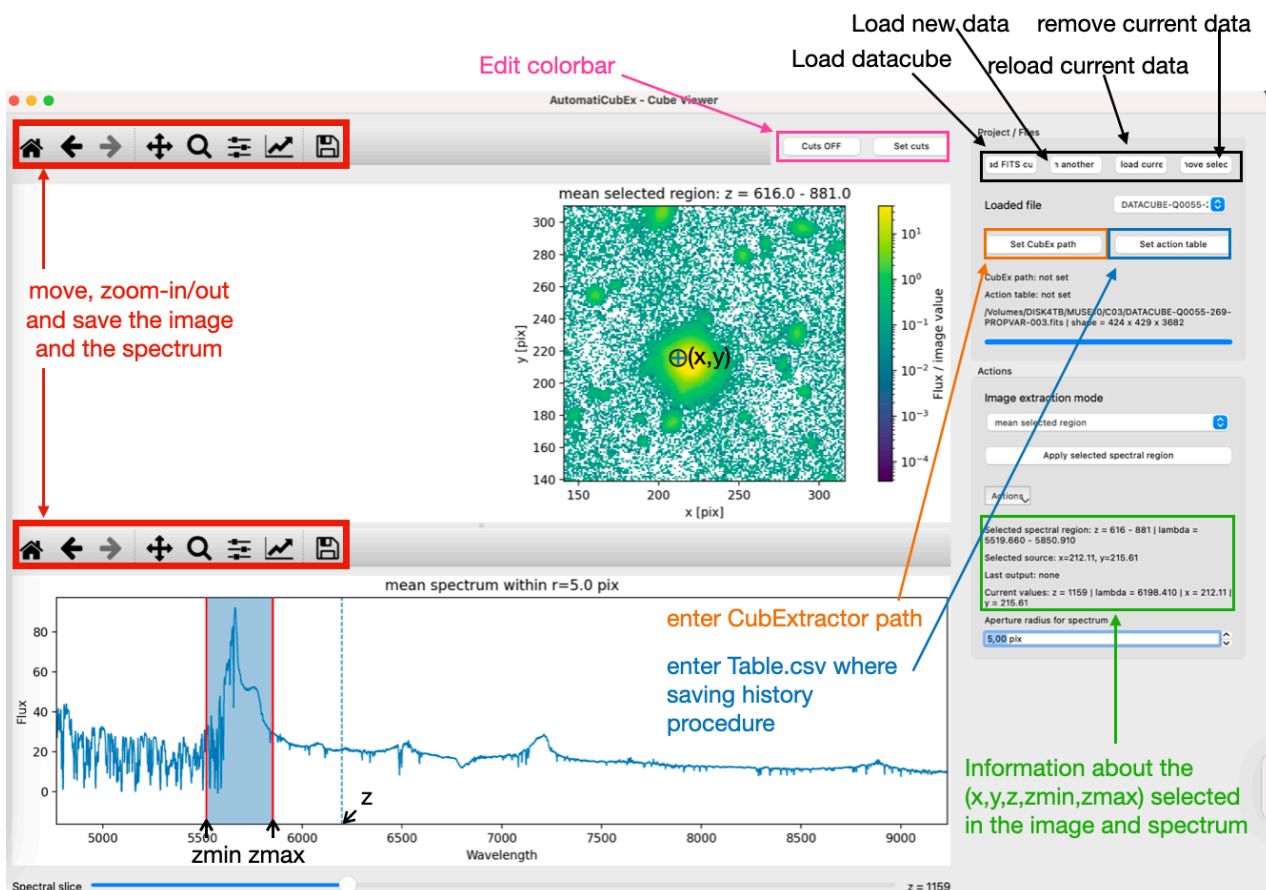
The “Actions” menu gives access to all CubEx-related processing tools, including:

- Cube2Im,
- CubePSFSub,
- CubeBKGSUB,
- CubeSel,
- CreateMask,
- CubEx,
- Binarize3DMask,
- CreateMaps.

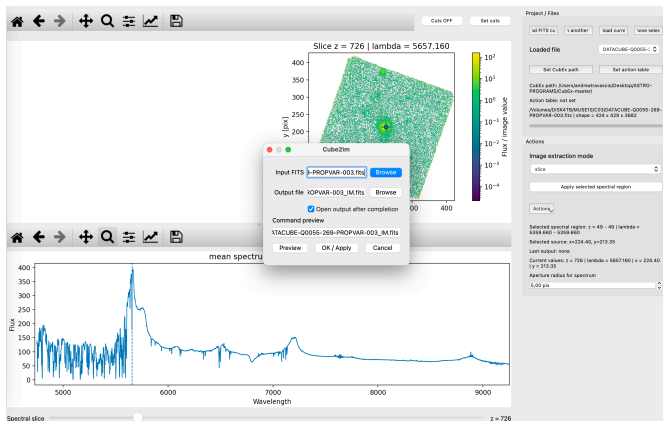
The green information box continuously updates the currently selected:

- spatial coordinates (x,y),
- spectral range (zmin,zmax),
- wavelength,
- aperture radius,
- current slice position.

Finally, the “Cuts OFF” and “Set cuts” buttons allow interactive control of the image color scale, similarly to ds9 or QFitsView. The user can dynamically modify the image contrast and colorbar range directly from the displayed image.

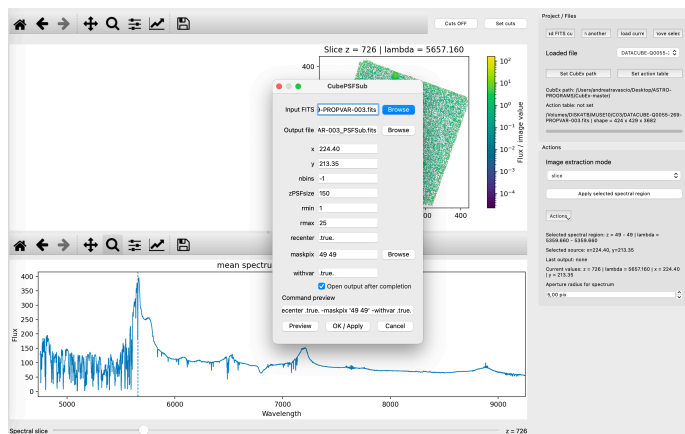


2. Generating a White-Light Image with Cube2Im



The first processing step is usually the generation of a white-light image (WLI) from the original datacube using Cube2Im. This image is useful for source identification and mask generation. Only the input cube and output image are required for this step.

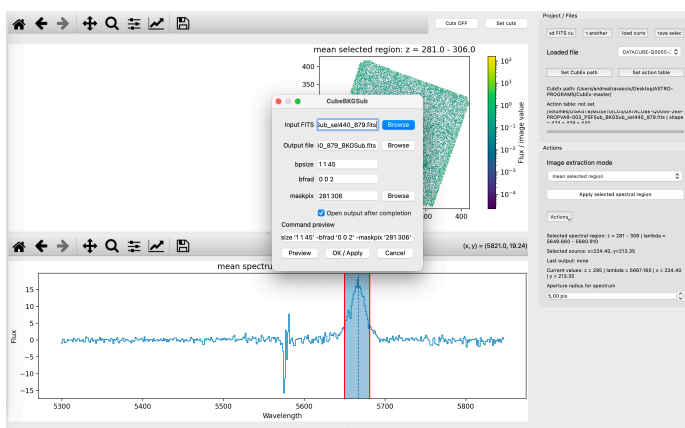
3. PSF Subtraction Workflow



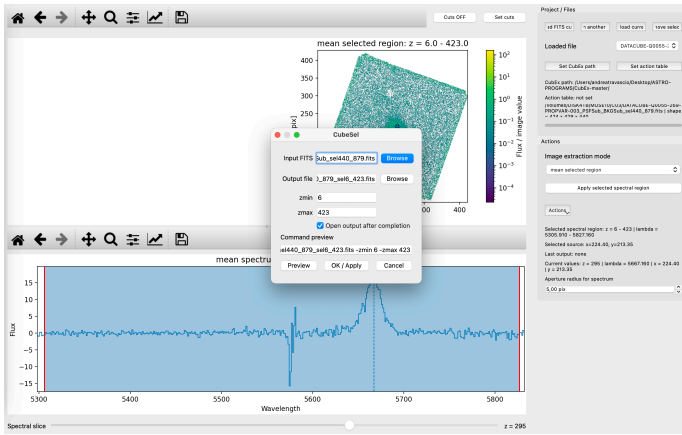
A standard analysis generally requires two PSF subtraction runs. First, the subtraction is performed without spectral masking in order to identify the approximate wavelength range of the extended emission. Afterwards, the same procedure is repeated while masking the spectral region containing the nebular emission. In AutomatiCubEx, this masking

region can be selected directly from the spectrum panel. The QSO centroid should be carefully selected before running the PSF subtraction. Parameters such as PSFsize and rmax strongly affect the quality of the empirical PSF model.

4. Background Subtraction and CubeSel

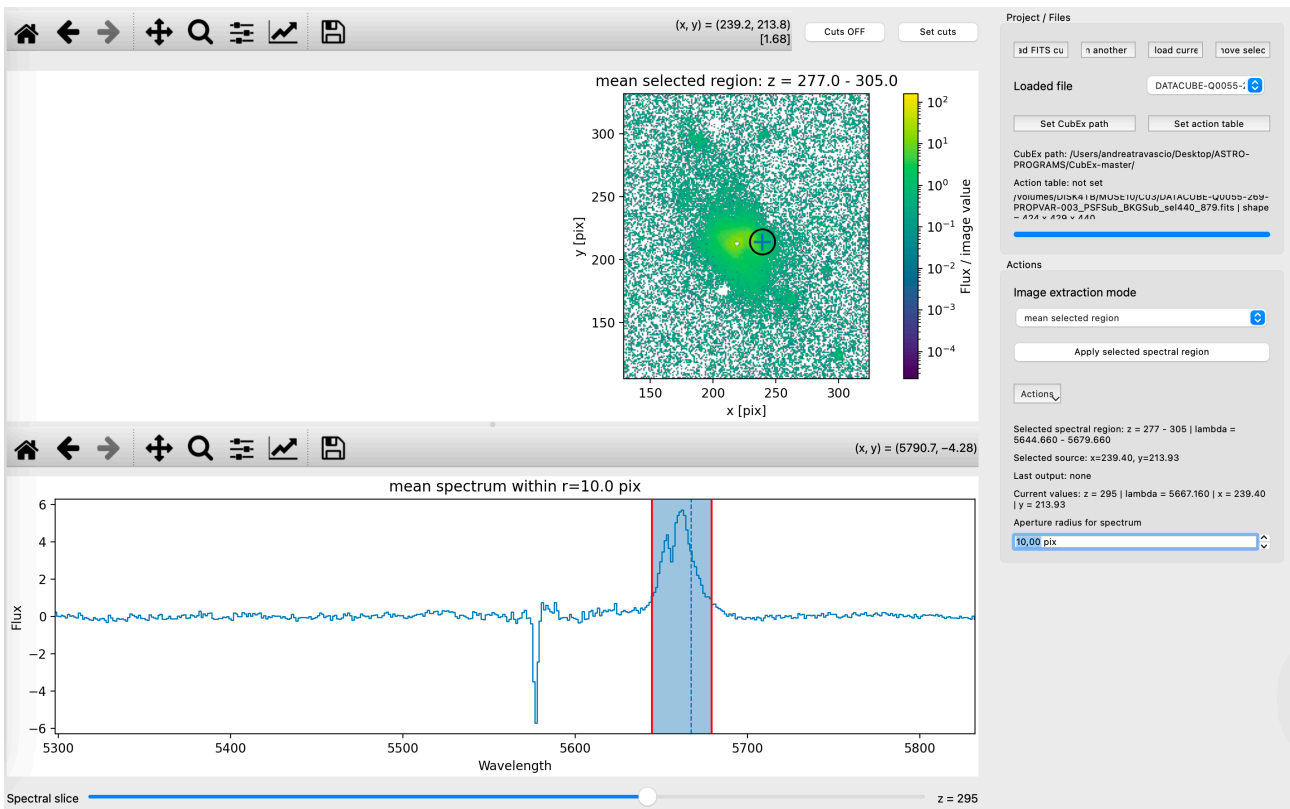


After PSF subtraction, the background subtraction is applied. The same spectral masking region should normally be used in order to avoid biasing the background estimate. CubeSel is then used to reduce the spectral size of the datacube, keeping only the wavelength region containing the emission line and a small continuum region around it.

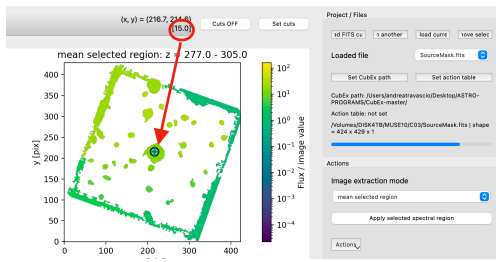


5. Interactive Exploration of the Emission

The GUI allows the user to select a spatial aperture and inspect the corresponding spectrum interactively. By selecting a spectral interval, the user can generate a mean or median image of the selected wavelength range. This is extremely useful for locating faint extended emission and determining the proper spectral range for extraction.

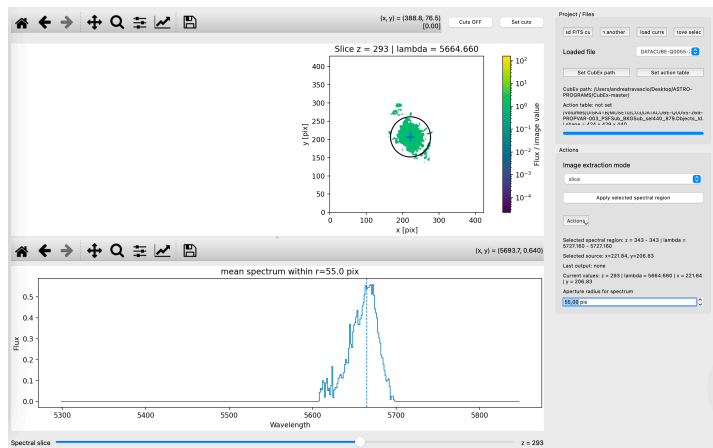
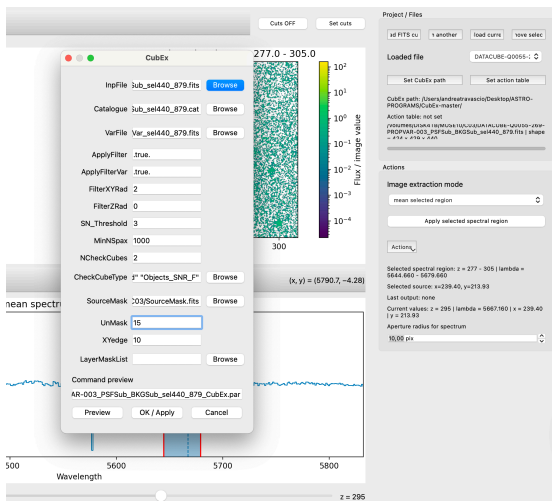


6. CreateMask and CubEx Extraction



CreateMask generates a source mask from the white-light image. Each source is assigned an ID number. The user should identify the source around which the extended emission is searched and remember its ID.

CubEx is then applied to the PSF-subtracted, background-subtracted, and CubeSel-cut datacube. Important parameters include: - SN_Threshold - MinNVox / MinNSpax - SourceMask - UnMask source ID CubEx produces a 3D segmentation mask identifying extended structures (right panel).



7. Binarize3DMask and CreateMaps

The Binarize3DMask tool allows the user to isolate one or multiple detected regions by converting selected IDs to 1 and all remaining voxels to 0. Finally, CreateMaps generates: - Surface brightness maps - Velocity maps - Kinematic products (right panel) using the final binary 3D mask.

